STAFF REVIEW · H446 006 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q4

Do Now — source answers

1. 5.5 Mantissa 01011000 = 0.6875 ; exponent 0011 = 3 ; $0.6875 \times 2^3 = 5.5$.

2. 01100000 1111 0.375 = $0.011_2 = 0.11_2 \times 2^{-1}$. Mantissa 01100000 ; exponent 1111 = −1 .

3. −10 −128 + 64 + 32 + 16 + 4 + 2 = −10 .

4. $AD_{16} = 10101101_2 \rightarrow 01010110_2 = 86$ A logical right shift inserts 0 on the left and discards the rightmost bit.

I Do — source worked steps

1. Use the common exponent 3 — Raise the second exponent from 1 to 3, so shift its mantissa right twice using sign extension. — CODE:
010100 0011
011000 0001 → 000110 0011

2. Use a temporary extra sign bit — CODE:
0010100
+ 0000110
= 0011010 — The working sum is $26/32 \times 2^3$.

3. Return to the requested width — CODE:
0011010
→ 011010 0011 — The mantissa already begins 01 , so the exponent stays at 3.

4. Verify in denary — CODE:
5 + 1.5 = 6.5

We Do — guided CHECK

1. Build the answer

2. Build the answer

3. Build the answer

4. Build the answer
Use the rounding or truncation rule specified by the question.

5. Build the answer

You Do — source mark schemes

1. Use common exponent 6 (0110). Shift the smaller-scale mantissa right, repeating its sign bit. CODE:
Aligned mantissas: 011000 and 000101
Temporary extra sign bit, same fractional places:

0011000
+ 0000101
= 0011101
The unnormalised sum is 29/32 × 2 6 . Normalise and return to the requested field widths.
CODE:
Final: 011101 0110
Denary check: 48 + (10) = 58.
Common exponent 6 (1), second mantissa 000101 (1), correct addition (1), final mantissa 011101 (1), exponent 0110 (1).

2. Use common exponent 3 (0011). Shift the smaller-scale mantissa right, repeating its sign bit.
CODE:
Aligned mantissas: 011000 and 001000
Temporary extra sign bit, same fractional places:
0011000
+ 0001000
= 0100000
The unnormalised sum is 32/32 × 2 3 . Normalise and return to the requested field widths.
CODE:
Final: 010000 0100
Denary check: 6 + (2) = 8.
Common exponent 3 (1), second mantissa 001000 (1), widened sum 0100000 (1), final mantissa 010000 (1), exponent 0100 (1).

3. Use common exponent 3 (0011). Shift the smaller-scale mantissa right, repeating its sign bit.
CODE:
Aligned mantissas: 101100 and 000110
Temporary extra sign bit, same fractional places:
1101100
+ 0000110
= 1110010
The unnormalised sum is -14/32 × 2 3 . Normalise and return to the requested field widths.
CODE:
Final: 100100 0010
Denary check: -5 + (1.5) = -3.5.
Common exponent 3 (1), second mantissa 000110 (1), signed sum −14/32 (1), normalised mantissa 100100 (1), exponent 0010 (1).

4. a) Matching positions must have the same scaled weight (1), so operands need the same exponent before direct addition (1). b) Repeat the sign bit / insert 1 (1), preserving the negative sign as the mantissa is halved (1). c) A short mantissa may discard nonzero low-order bits during alignment or final storage (1).

Exit Ticket — source answer

1. True. Align the exponents first so the mantissas represent values at the same scale.